This handout includes space for every question that requires a written response. Please feel free to use it to handwrite your solutions (legibly, please). If you choose to typeset your solutions, the —README.md— for this assignment includes instructions to regenerate this handout with your typeset LaTeX solutions.

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# 1.a

It's generally very difficult to translate high level objectives into a math mathematical reward function an agent can optimize and it is very difficult to avoid reward hacking that results in an unintended consequences.

Example: Content recommendation agent

The reward function could be number of clicks. The unintended result could be recommending all the clickbait contents instead of high quality, actually interesting contents.

# 1.b

It's a single legged robot trying to hop forward.

$$\text{reward} = \text{healthy\_reward} + \text{forward\_reward} - \text{ctrl\_cost}$$

Where

$$\begin{aligned} \text{healthy\_reward} &: \text{Keep the agent to stand by keeping the torso in a healthy state} \\ \text{forward\_reward} &: \text{Push the agent to move forward by maximize the x-velocity} \\ \text{ctrl\_cost} &: \text{Prevent the agent to aggressively move, i.e. move smoothly by penalize the large control} \end{aligned}$$

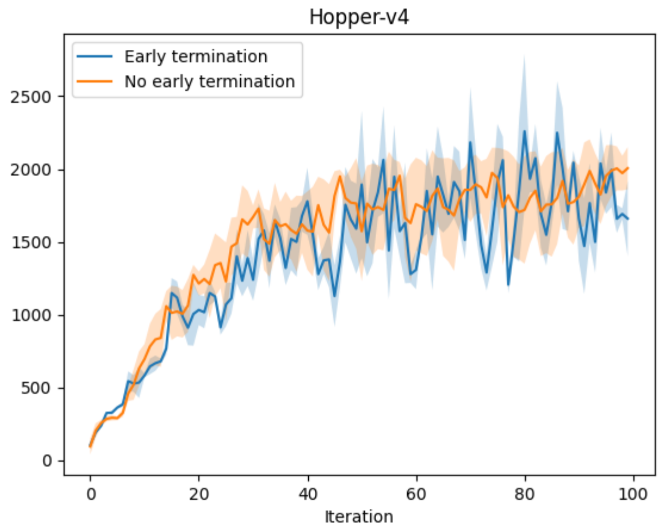
# 1.c

It means the hopper is standing and not falling over. It is determine by two factors: the torso's height must be above 0.8 meters, and its angle must be within 0.2 radians of vertical.

Advantage of early-termination: speed up training.

Disadvantage of early-termination: can't learn to recover from falling.

# 1.e




The performance without early-termination is better than with early-termination. It also more consistent, i.e. the standard error is smaller. To get better estimate, we should sample more times by using different random seeds (at least 10-20 times).

# 1.f

I looked at a rendered video of the agent trained without early-termination.

The agent seem to learn some basic control an able to move forward. But it also keep repeatedly falling and trying to stand up again.

It partially complete the task because it's moving. But it fails to keep the agent in a healthy state. It doesn't seem to be too surprised because the reward function doesn't care about the continuous time it keep in the healthy state.

# 1.g

I looked at another without-early-termination model.

The agent quickly falling over with very little progress. It still able to crawling forward on the ground but never stand up again. I would consider this a completely failing.

# 2.a

Let $R_1 = \sum \hat{r}(o^1, a^1) = \sum \phi_w(o^1, a^1)$ and $R_2 = \sum \hat{r}(o^2, a^2) = \sum \phi_w(o^2, a^2)$.
We can express the Bradley-Terry in sigmoid form.

$$\begin{aligned}
\hat{P}[\sigma^1 > \sigma^2] &= \frac{e^{R1}}{e^{R1} + e^{R2}} \\
&= \frac{1}{1 + e^{-(R1-R2)}} \\
&= \sigma(R_1 - R_2)
\end{aligned}$$

Substitute into loss function

$$L = -\left(\mu(1)\log(\sigma(R_1 - R_2)) + \mu(2)\log(\sigma(R_2 - R_1))\right)$$

Let $z = R_1 - R_2$, applying chain rule

$$\begin{aligned}
\nabla_w L &= [-\mu(1)(1 - \sigma(z)) + \mu(2)\sigma(z)]\nabla_w z \\
&= [-\mu(1) + \mu(1)\sigma(z) + \mu(2)\sigma(z)]\nabla_w z \\
&= [(\mu(1) + \mu(2))\sigma(z) - \mu(1)]\nabla_w z \\
&= [(\mu(1) + \mu(2))\sigma(R_1 - R_2) - \mu(1)]\nabla_w (R_1 - R_2) \\
&= [(\mu(1) + \mu(2))\sigma(\sum_t \phi_w(o_t^1, a_t^1) - \sum_t \phi_w(o_t^2, a_t^2)) - \mu(1)](\sum_t \nabla_w \phi_w(o_t^1, a_t^1) - \sum_t \nabla_w \phi_w(o_t^2, a_t^2))
\end{aligned}$$

# 2.b

I got the following pairs

3147 (equally preferred): agree, no obvious differences

## 2.c

Four more pairs:

4191 (equally preferred): agree, no obvious differences

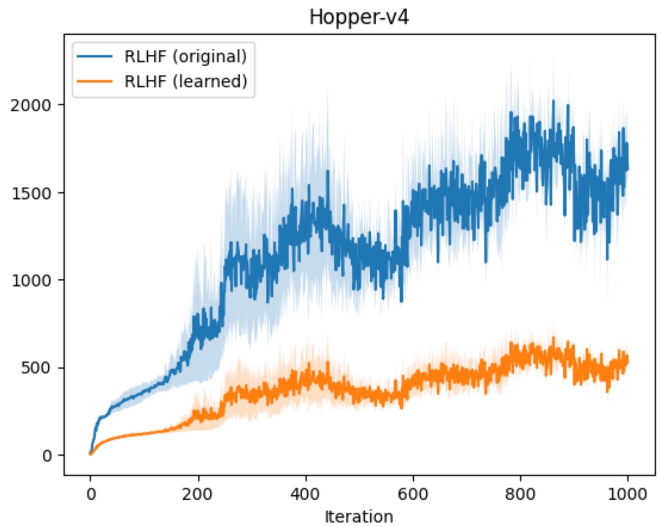
6599 (one is preferred): not agree, the non-preferred one has better ending ankle angle to land

6844 (one is preferred): similar to 6599

7324 (one is preferred): agree, the preferred one landed on its heel, which is a better position to initiate the next hop

I am not confident to say I trust this data because firstly the sample is very short, and I disagree some of the samples.

# 2.e




They are highly correlated.

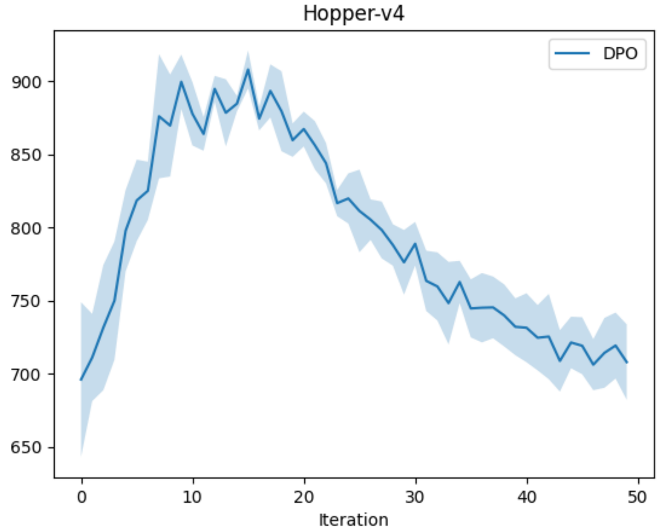
## 2.f

It does not identifiable because it has very similar trend as the original reward.

# 2.g

It can perform more hops but fall and never get up again. I think it is better in terms of hopping but the original policies can move forward more. Can't say which one is better but the RLHF seems working as we prefer a landing position that is easier to initiate next hop.

# 3.d




The DPO reaches max reward of 950, which better than RLHF but it degraded after it reaches the peak, suggesting overfitting.

Pros and Cons

| Method | Pros | Cons |
|---|---|---|
| RLHF | More intuitive as it has a two-staged structure. | Unstable. The error in the reward function can be exaggerated by the RL algorithm. |
| DPO | Avoiding a separate reward function.<br>The RL algorithm can directly optimize the policy. | May overfit the preference data. |

## 3.e

To be honest, I don't see significant differences. DPO doesn't seem to improve the performance.

# 4.a

The raters are often contract worker or from a crowdsourcing platform. These worker typically are from a low-income background. If the raters are not diverse enough, the model may has strong bias on some sensitive topics.

# 4.b

No, the medical diagnoses is strongly rely on domain knowledge and must be objective. It shouldn't rely on preferences. Unless the model has acceptable level of correctness, the RLHF may be used to improve the way it structure the response.

# 4.c

RLHF is lack of factuality checks. Human raters may be lack of specific domain knowledge to do factuality checks and prefer answers that is well-written or easier to read, even if they are inaccurate.